University College Roosevelt
Lange Noordstraat 1, Middelburg, The Netherlands
Bachelor of Liberal Arts & Sciences

Senior Project

Comparing classical and neural collaborative filtering models on implicit feedback data

Comparing collaborative filtering models

Author:
G.V. Vetsler
9783954 (Student Number)

Supervisor:
Dr. Marijn Jansen

Second examiner:
Dr. Tamas Vinko

Table of contents

1. Motivation and problem setting
2. Data and evaluation protocol
3. Models (classical and neural)
4. Results (MovieLens vs Yelp)
5. Conclusions, limitations

The problem: recommending from implicit feedback

Real systems: Netflix, Spotify, TikTok, Yelp, e-commerce.
They observe mostly implicit signals: clicks, views, purchases.
Key challenge: "No interaction" does not mean dislike.
So we treat recommendation as a ranking problem: what should be top for each user?

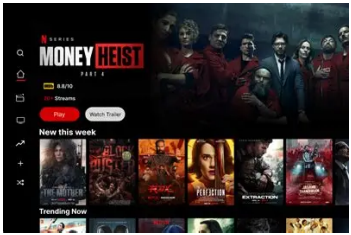


Figure: Netflix recommendation page. Source: Figma Community — Netflix Home Page (Desktop & TV).

Collaborative Filtering in one slide

Core idea: recommend items using *other users' behavior*, not item content.

Users	Movie 1	Movie 2	Movie 3	Movie 4
User 1	5	4		5
User 2	4		3	
User 3		1		2
User 4	1	2		

Figure: Example of the Collaborative filtering idea

If User 1 and User 2 both liked Movies 1 and 2, and User 1 also liked Movie 4, then Movie 4 becomes a good candidate recommendation for User 2.

Datasets

- MovieLens-1M (movies): denser interactions, Movies vs Users
- Yelp (businesses): much sparser, local/business structure, geographical semantics

MovieLens-1M

Users (M)	6,040
Items (N)	3,706
Interactions	1,000,209
Density	4.4684%
Sparsity	95.5316%
Avg. / user	165.60
Avg. / item	269.89

Yelp

Users (M)	24,930
Items (N)	25,799
Interactions	690,381
Density	0.1073%
Sparsity	99.8927%
Avg. / user	27.69
Avg. / item	26.76

Models compared

Classic baselines:

- ItemKNN: recommend items similar to what you liked

MF models:

- MF-BPR: matrix factorization trained for ranking (BPR)
- GMF: shallow neural MF variant

Neural CF family:

- MLP: non-linear interaction function
- NeuMF: GMF + MLP hybrid (with and without pretraining)

ConvNCF:

- outer product $\rightarrow$ interaction map $\rightarrow$ CNN

Matrix Factorization (MF): the classic baseline

Idea: represent each user and item with a low-dimensional vector (embedding).

- User u has embedding $\mathbf{p}_u \in \mathbb{R}^k$, item i has embedding $\mathbf{q}_i \in \mathbb{R}^k$.
- Predicted preference score:

$$\hat{y}_{ui} = \mathbf{p}_u^\top \mathbf{q}_i$$

- Interpretation: if vectors align, the score is high (good match).
- Limitation: interaction is linear (dot product) $\rightarrow$ cannot directly model complex non-linear patterns.

Link to GMF

GMF is a neural generalization of MF: it replaces the fixed dot product with a learnable weighted element-wise interaction, while staying almost as simple.

MF-BPR

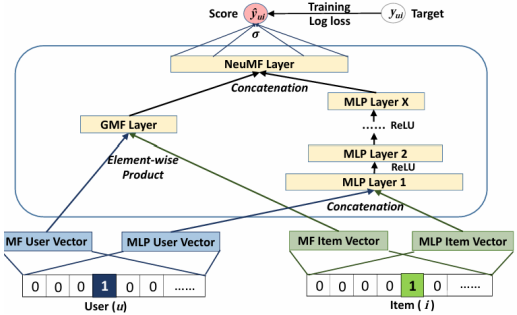
Same MF scoring function, but trained to rank observed items above unobserved ones using BPR loss.

NeuMF in one picture (hybrid model)

Idea: keep MF-like interaction *and* add non-linear capacity.

- GMF branch: element-wise product
- MLP branch: concatenation + dense layers
- Fuse at output

Pretraining variant: train GMF and MLP first, then use their weights to initialize NeuMF.



Schematic illustration of the NeuMF architecture combining GMF and MLP.(Figure adapted from [3]).

Why CNN shows up in recommendation (ConvNCF intuition)

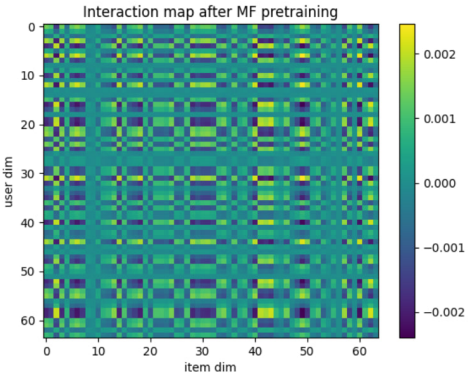
CNN = processes images / 2D patterns.
ConvNCF creates a 2D "image" from embeddings:

$$\mathbf{E}_{ui} = \mathbf{p}_u \mathbf{q}_i^\top \in \mathbb{R}^{k \times k}$$

Claim: this interaction map contains local patterns a CNN can detect.

Key skepticism

Unlike pixels in images, embedding dimensions have no natural spatial order.



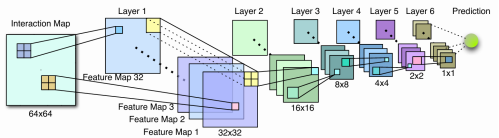
Interaction map (outer product of embeddings) after MF pretraining (MovieLens)

ConvNCF architecture (outer product → CNN)

Pipeline:

1. user/item embeddings
2. outer product → $k \times k$ interaction map
3. several Conv2D layers
4. flatten + output

In this thesis: a shallow ConvNCF variant: {32,32,64} filters, (compute constraints).



An example of the architecture of our ConvNCF model that has 6 convolution layers with embedding size 64 (Figure adapted from [2]).

Training setup (high level)

Negative sampling: for each positive interaction, sample unobserved items as negatives.
Two training objectives:

- Pointwise (GMF / MLP / NeuMF): classify observed vs sampled unobserved (negative)
- Pairwise BPR (MF-BPR / ConvNCF): rank positive above negative

Hyperparameter search: For NeuMF and ConvNCF models a limited hyperparameter search was conducted
Best weights search: Using validation dataset, pick the best weights.

Evaluation protocol

Leave-one-out:

- For each user, hold out the most recent interaction as test
- Add 99 randomly sampled negatives for evaluation
- Validate using the same protocol on the next most recent interaction
- Train on the remaining interactions

Ranking metrics at $K = 10$:
Predict 10 items, per user:

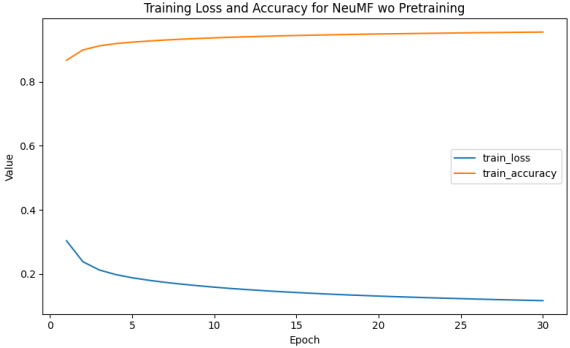
- HR@10: how many true items are in those Top-10?
- NDCG@10: rewards placing the true item higher in the Top-10

Research questions

1. Are MLP-based models a good fit for CF?
2. Is ConvNCF a good fit and the outer product actually has spatial features that can be captured by CNNs?
3. To what extent deep learning is helpful in the CF models?

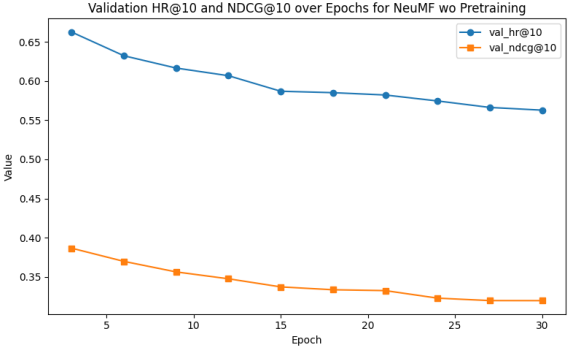
Training dynamics: NeuMF on MovieLens

NeuMF training (loss + accuracy)



NeuMF on MovieLens: training loss and training accuracy over epochs.

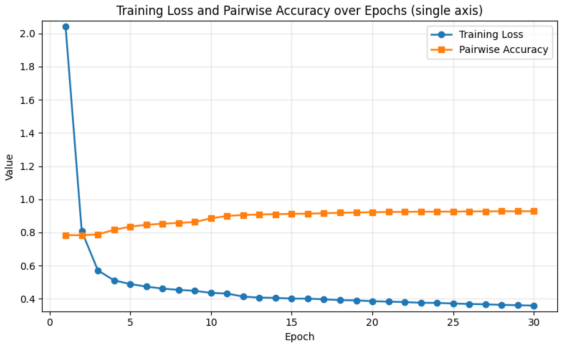
NeuMF validation (HR@10 + NDCG@10)



NeuMF on MovieLens: validation HR@10 and NDCG@10 (evaluated every 3 epochs).

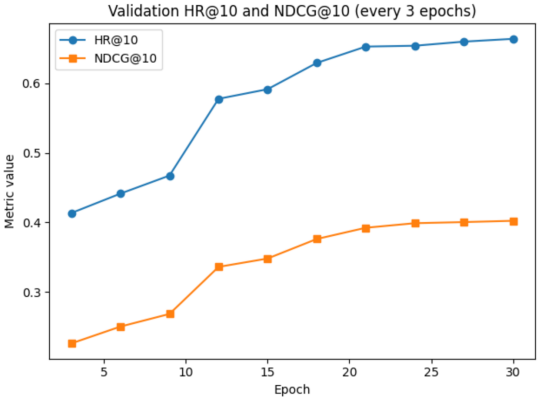
Training dynamics: ConvNCF on Yelp

ConvNCF training (loss + accuracy)



ConvNCF on Yelp: training loss and training accuracy over epochs.

ConvNCF validation (HR@10 + NDCG@10)



ConvNCF on Yelp: validation HR@10 and NDCG@10 (evaluated every 3 epochs).

Results on MovieLens-1M (top-10 accuracy)

Metric focus: HR@10 and NDCG@10 (higher is better)

Model	HR@10	NDCG@10
GMF	**0.6629**	**0.3853**
NeuMF-pre	0.6480	0.3814
NeuMF	0.6457	0.3752
MLP	0.6358	0.3675
ItemKNN	0.5204	0.3022
MF-BPR	0.5379	0.2989
ConvNCF	0.3220	0.1606

Interpretation

On MovieLens, the simpler embedding-based model (GMF) is best.
ConvNCF fails badly: high complexity, low generalization.

Results on Yelp (top-10 accuracy)

Metric focus: HR@10 and NDCG@10

Model	HR@10	NDCG@10
ItemKNN	**0.7301**	**0.4747**
NeuMF-pre	0.6917	0.4297
MLP	0.6787	0.4224
GMF	0.6775	0.4284
NeuMF	0.6744	0.4264
ConvNCF	0.6242	0.3702
MF-BPR	0.4329	0.2666

Interpretation

On Yelp, a classic baseline (ItemKNN) wins.
Neural MF-style models are competitive but do not dominate.

RQ1: Are MLP-based models a good fit for CF?

- MLP-based models (MLP, NeuMF) are competitive, but not consistently better than simpler MF-style models.
- On MovieLens, GMF achieves the best ranking quality (highest NDCG@10); NeuMF variants are close but not dominant.
- Validation curves often peak early and then decline $\rightarrow$ higher capacity can translate into overfitting.
- Pretraining helps NeuMF, but the improvement is modest.
- This matches prior findings that strong MF-style baselines remain highly competitive and that gains from neural CF can be small under fair evaluation [4, 1].

Takeaway

More non-linearity does not automatically translate into better top-K recommendations.

RQ2: Is ConvNCF a good fit, and does the outer product have CNN-friendly patterns?

- MovieLens: ConvNCF generalizes very poorly (low and unstable validation metrics), despite fitting training data well.
- Yelp: ConvNCF works better than on MovieLens, but still lags behind strong baselines and MF-style neural models.
- Core concern: CNNs assume spatial locality, but embedding dimensions have no natural order.
- Even where ConvNCF improves (Yelp), a simple ItemKNN baseline remains stronger.
- This aligns with replication-based critiques arguing that CNN locality assumptions do not naturally apply to embedding interaction maps [1].

Takeaway

The CNN inductive bias does not reliably match the structure of outer-product interaction maps.

RQ3: How much does deep learning help vs well-tuned classical baselines?

- On MovieLens, embedding-based models (especially GMF) clearly beat classical baselines (KNN, MF-BPR).
- Extra architectural complexity (MLP depth, CNNs) yields small or negative gains under limited tuning.
- On Yelp, strong neighbourhood structure makes ItemKNN the best accuracy model.
- NeuMF-pre is robust and competitive, but not a universal winner across both dataset types.
- Overall, this supports the broader message that careful baselines and fair evaluation are essential when claiming neural CF improvements [4, 1].

Takeaway

Deep learning helps mainly via good embeddings and training, not automatically via complexity.

Limitations

- **Restricted hyperparameter search:** deep models are sensitive (LR, L2, batch size, sampler).
- **Training horizon:** some models improve with longer training (ConvNCF in original work trains very long).
- **ConvNCF variant:** shallow CNN used (stability/compute constraints).
- **Negative sampling:** simple uniform sampling; smarter/harder negatives may change outcomes.

Conclusion

Under controlled conditions and limited tuning budget:

- **NeuMF** is competitive, but does not consistently beat simpler models.
- **GMF** (shallow) is best on MovieLens.
- **ItemKNN** is best on Yelp.
- **ConvNCF** does not justify its complexity in this setup.

Acknowledgments

- I would like to thank Dr. Marijn Jansen for supervision, guidance, and feedback throughout this project.
- Thank you for your attention.

References I

Vito Walter Anelli, Alejandro Bellogín, Tommaso Di Noia, Eugenio Di Sciascio, and Francesco Narducci.
Reenvisioning the comparison between neural collaborative filtering and matrix factorization.
In *Proceedings of the Fifteenth ACM Conference on Recommender Systems (RecSys 2021)*. ACM, 2021.

Xiangnan He, Xiaoyu Du, Xiang Wang, Feng Tian, Jinhui Tang, and Tat-Seng Chua.
Outer product-based neural collaborative filtering.
In *Proceedings of the 27th International Joint Conference on Artificial Intelligence (IJCAI-18)*, pages 2227–2233, 2018.

References II

Xiangnan He, Lizi Liao, Hanwang Zhang, Liqiang Nie, Xia Hu, and Tat-Seng Chua.
Neural collaborative filtering.
In *Proceedings of the 26th International Conference on World Wide Web (WWW '17)*, pages 173–182. ACM, 2017.

Steffen Rendle, Walid Krichene, Li Zhang, and John Anderson.
Neural collaborative filtering vs. matrix factorization revisited.
In *Proceedings of the Fourteenth ACM Conference on Recommender Systems (RecSys 2020)*, pages 240–248. ACM, 2020.